IT342 – System Integration and Architecture

MedBuddy

Project Development Phase 2

Mobile Development

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Section	IT342-G2
Project Name	MedBuddy
Date	March 30, 2026
Phase	Phase 2 – Mobile Authentication
Status	Completed

1. Final Commit Information

Commit Message

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IT342 Phase 2 - Mobile Development Completed
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Repository Details

Repository Name: IT342_MedBuddy_Aquino

Branch: main

Commit Hash: ab0a90357cb6426fe86eb897d4c72e1ab7461aa9

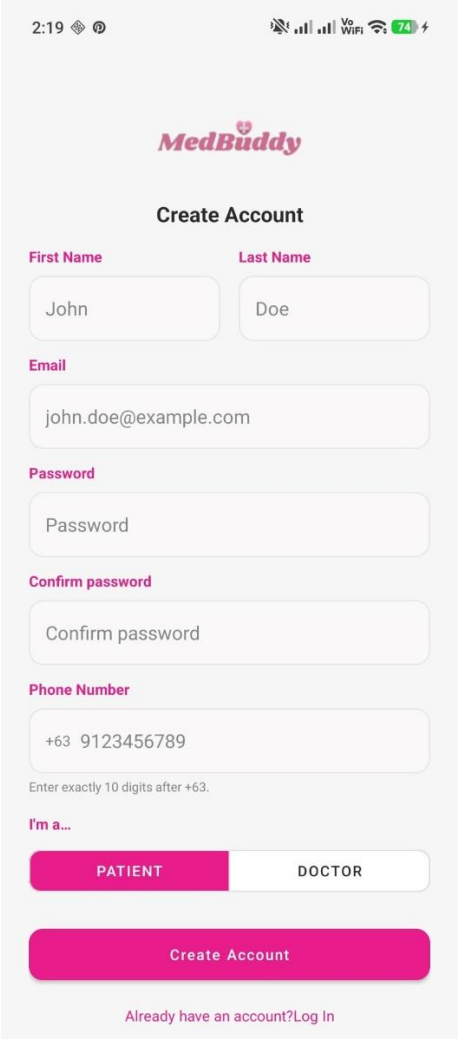
Commit Link: https://github.com/oblivjules/IT342_MedBuddy_Aquino/commit/ab0a90357cb6426fe86eb897d4c72e1ab7461aa9

Repository Link: https://github.com/oblivjules/IT342_MedBuddy_Aquino.git

2. Screenshots of Implementation

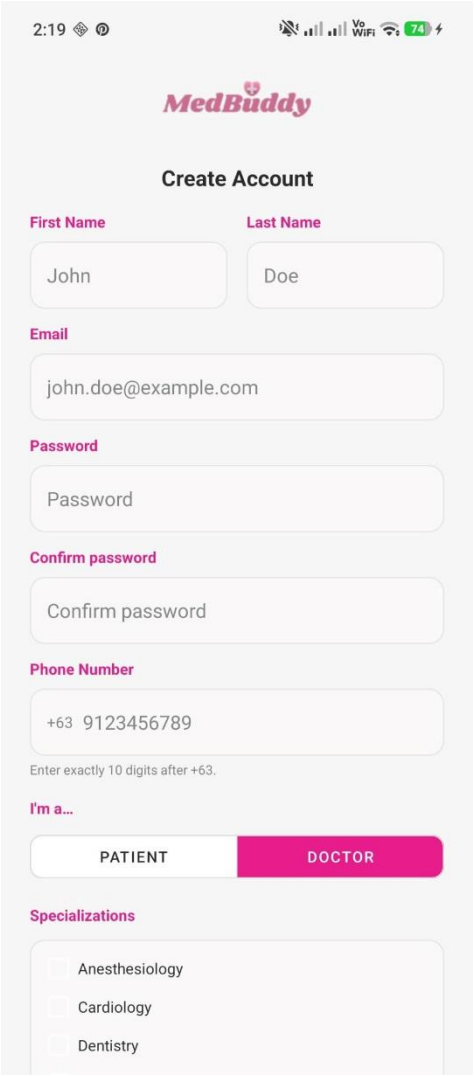
Screenshot 1 – Registration Page

a. Patient Registration



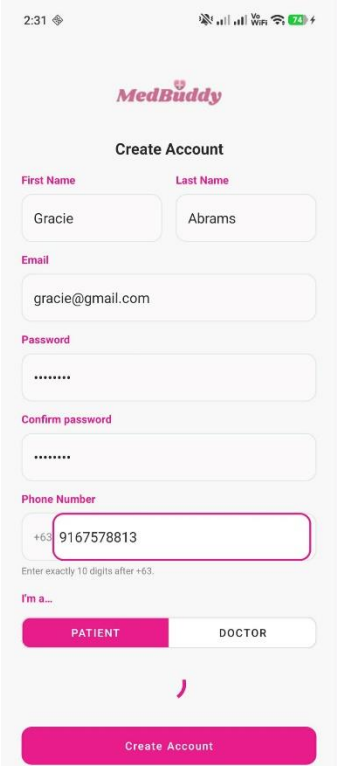
2. Screenshots of Implementation

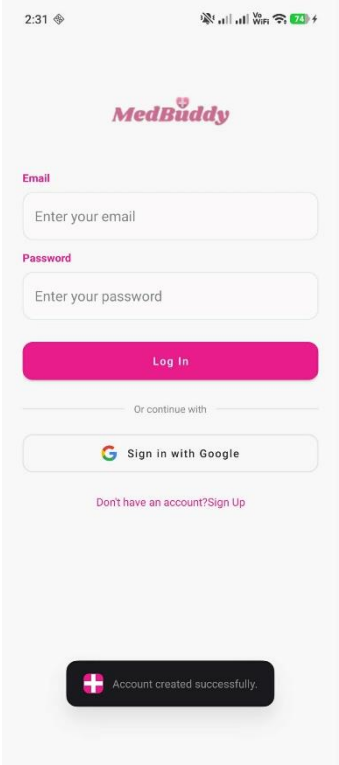
b. Doctor Registration



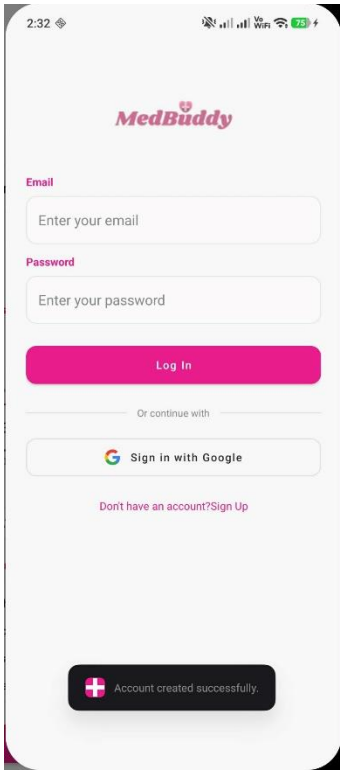
Screenshot 2 – Successful User Registration

a. Patient

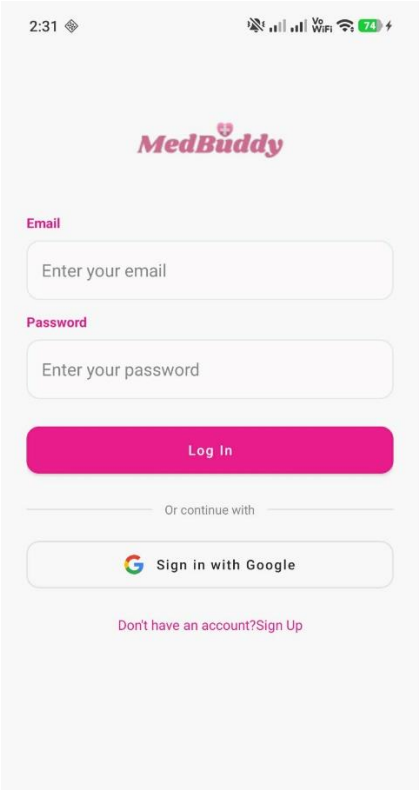




b. Doctor

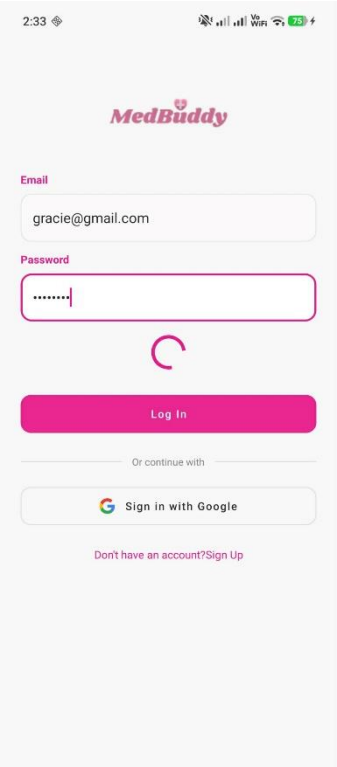


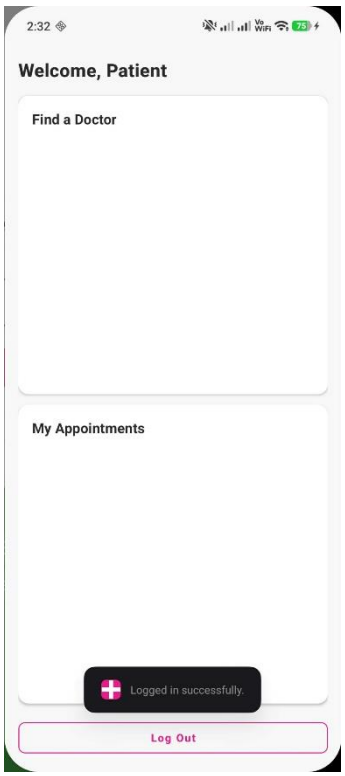
Screenshot 3 – Login Page (Same Login Page for both Roles)



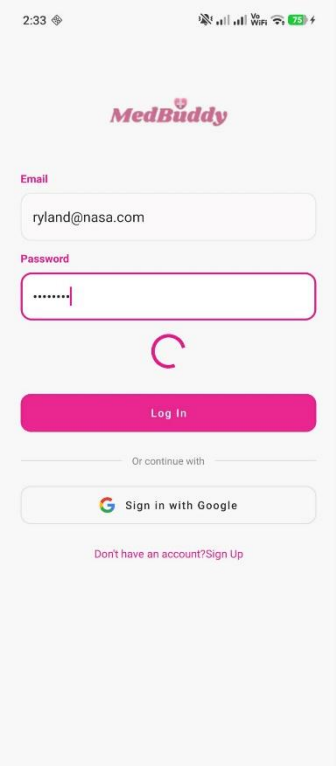
Screenshot 4 – Successful Login

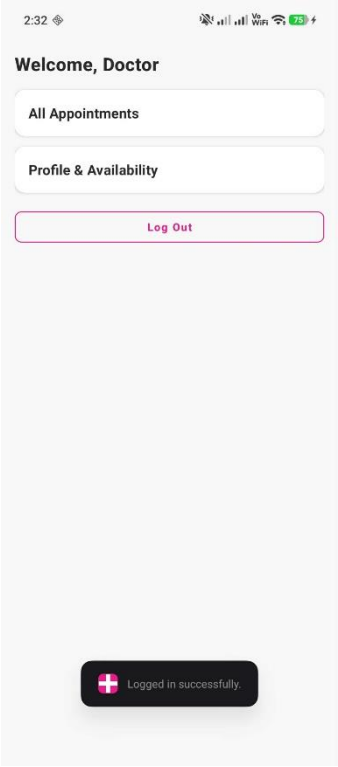
a. Patient





b. Doctor





Screenshot 5 – Database Record of Registered User

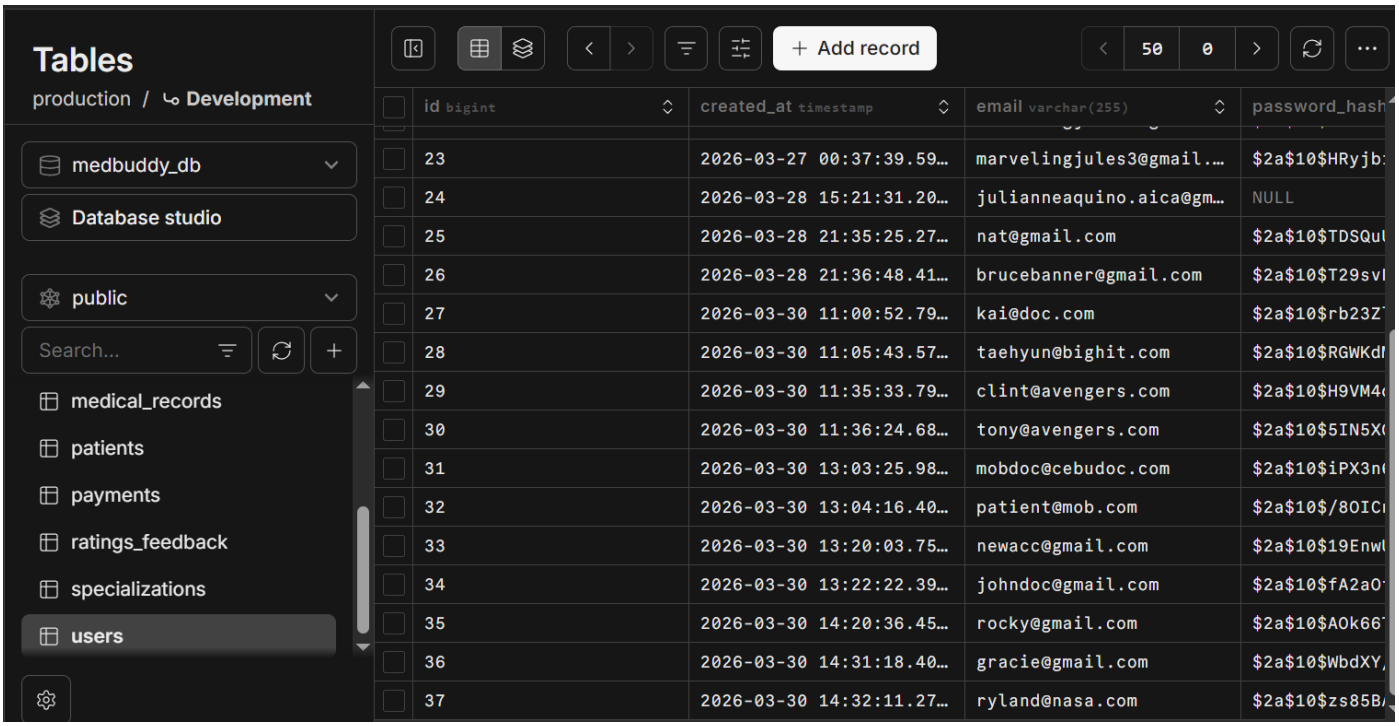
a. New Patient Record

22	Gracie	Abrams	+639167578813

b. New Doctor Record

15	Ryland	Grace	+639613784815

c. Both records showing on the Users Table



3. Implementation Summary

How Registration works:

- In RegisterActivity, user inputs are validated locally (required names, valid email, password length >= 8, password confirmation, and 10-digit phone).
- User chooses role (PATIENT or DOCTOR); if doctor is selected, specializations are fetched from GET /api/specializations and at least one must be selected.
- On submit, app calls POST /api/auth/register with RegisterRequest (including specializationIds for doctors and phone formatted as +63...).
- On success, it shows a success toast and redirects to LoginActivity; on failure, it parses server error (detail/message) and displays it.

How Login works:

- In LoginActivity, if a token already exists in TokenManager, it skips login and navigates straight to dashboard.
- Otherwise, it validates email/password are not blank, then calls POST /api/auth/login with LoginRequest.
- Response (AuthResponseDto) includes JWT token + user; both are saved via TokenManager (encrypted prefs if available, fallback to normal shared prefs).
- App routes user to DoctorDashboardActivity or PatientDashboardActivity based on role in the returned/saved user payload.

API integration used:

- Retrofit + Gson + OkHttp are used (mobile/app/build.gradle).
- Base URL comes from BuildConfig.API_BASE_URL (AppConstants.apiBaseUrl()), with build-type config:
    - debug default: http://10.0.2.2:8080/
- RetrofitClient adds Authorization: Bearer <token> automatically for protected endpoints, but skips auth header for public paths (/api/auth/login, /api/auth/register, /api/auth/health, /api/specializations).
- API contracts are defined in ApiService using suspend Retrofit endpoints (auth, users, doctors, specializations, appointments, availability).